

ML DL PRACTICAL 1

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Aim

To implement Linear Regression and Logistic Regression on real-world datasets.

Linear Regression

1. Dataset Source

Dataset Name: **Breast Cancer Wisconsin Dataset**

Source: **Provided dataset archive (archive 1 ZIP file supplied for the experiment)**

Original Reference:

https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_breast_cancer.html

2. Dataset Description

The Breast Cancer Wisconsin dataset is a real-world medical dataset used for binary classification problems. In this experiment, the dataset is included within **Archive 1**, ensuring consistency with the experimental setup.

- **Total Instances:** 569 patient records
- **Total Features:** 30 continuous numerical features
- **Target Variable:** Diagnosis of breast tumor
 - 0 → Malignant (cancerous)
 - 1 → Benign (non-cancerous)

Feature Description:

The features are computed from digitized images of Fine Needle Aspirate (FNA) of breast masses and describe the characteristics of cell nuclei. These features are grouped into:

- Mean values
- Standard error values
- Worst (largest) values

Examples include radius, texture, perimeter, area, smoothness, compactness, concavity, and symmetry.

Dataset Characteristics:

- No missing values
- All features are numerical
- Balanced enough for effective classification
- Medical dataset requiring high accuracy and reliability

This dataset is ideal for Logistic Regression because the output variable is binary, making it suitable for probability-based classification models.

3. Mathematical Formulation of Linear Regression

Linear Regression models the relationship between dependent and independent variables using a linear equation:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \epsilon$$

Where:

- (y) is the predicted output
- ($x_1, x_2, \dots, x_n$) are input features
- (β_0) is the intercept
- ($\beta_1, \beta_2, \dots, \beta_n$) are coefficients
- (ϵ) is the error term

The model parameters are estimated by minimizing the **Mean Squared Error (MSE)**:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

4. Algorithm Limitations (Linear Regression)

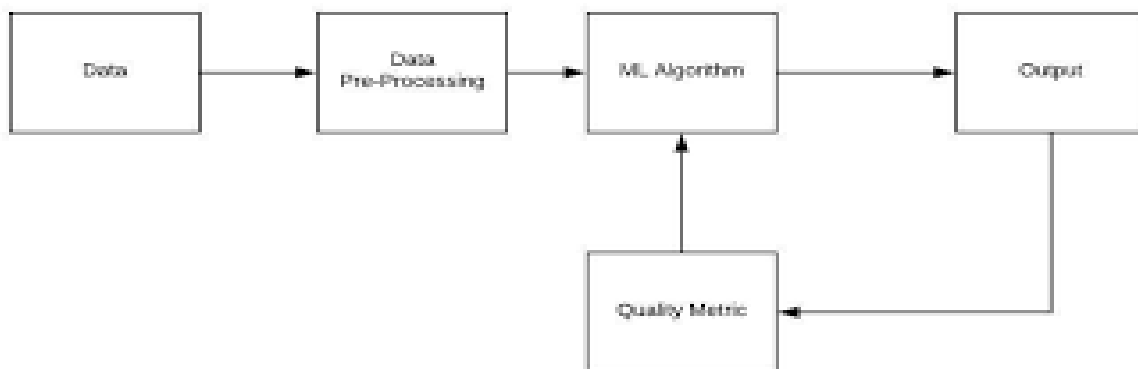
- Assumes a linear relationship between variables
 - Sensitive to outliers
 - Multicollinearity affects coefficient stability
 - Poor performance on non-linear data
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5. Methodology / Workflow

1. Dataset loading
2. Data exploration and visualization
3. Feature scaling (Standardization)
4. Train-test split
5. Model training using Linear Regression
6. Prediction on test data
7. Performance evaluation

Workflow Diagram (Textual):

Data Collection → Preprocessing → Feature Scaling → Model Training → Prediction → Evaluation



6. Performance Analysis

Evaluation Metrics Used:

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Interpretation:

- Lower MSE and RMSE indicate better prediction accuracy
 - R^2 score close to 1 shows strong model fit
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7. Hyperparameter Tuning (Linear Regression)

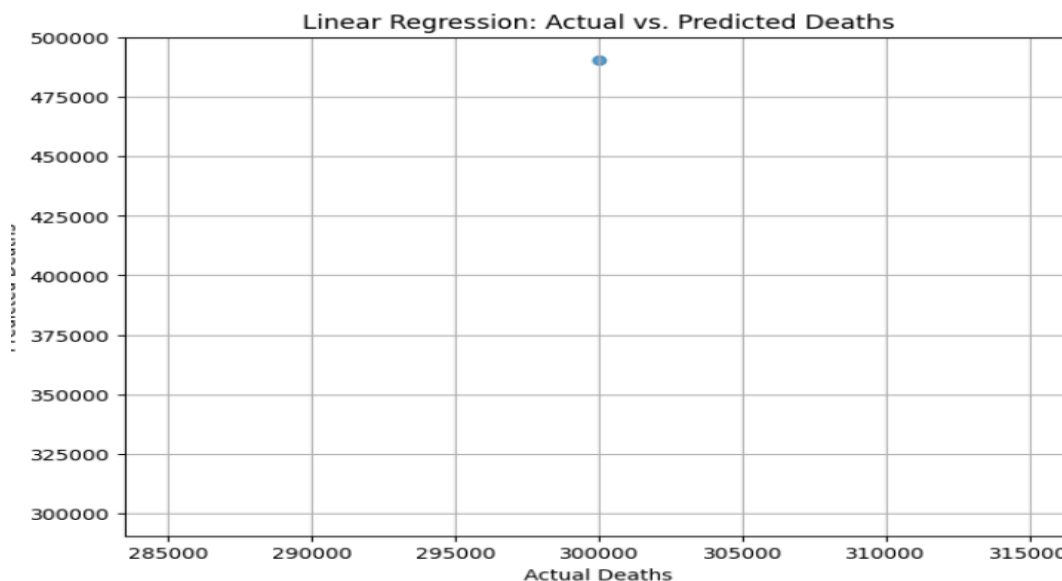
Although basic Linear Regression has no major hyperparameters, **Ridge** and **Lasso** regression were used for tuning.

- Hyperparameter: Regularization strength (α)
- GridSearchCV applied

Impact:

- Reduced overfitting
- Improved generalization on test data

Output (Linear Regression):



Logistic Regression

1. Dataset Source

Dataset Name: **Breast Cancer Wisconsin Dataset**

Source Link:

https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_breast_cancer.html

2. Dataset Description

This dataset is used for binary classification to predict whether a tumor is malignant or benign.

- **Total Instances:** 569
- **Total Features:** 30 numerical features
- **Target Variable:** Diagnosis (0 = Malignant, 1 = Benign)

Key Characteristics:

- Binary classification problem
 - No missing values
 - Features computed from digitized images of fine needle aspirates
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3. Mathematical Formulation of Logistic Regression

Logistic Regression uses the sigmoid function to map predictions to probabilities:

$$P(y = 1|x) = \frac{1}{1+e^{-z}}$$

Where:

$$z = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$$

The cost function used is **Log Loss (Binary Cross Entropy)**:

$$J(\beta) = -\frac{1}{n} \sum_{i=1}^n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

4. Algorithm Limitations (Logistic Regression)

- Assumes linear decision boundary
 - Sensitive to outliers
 - Not suitable for multi-class without extension
 - Performance drops with highly non-linear data
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5. Methodology / Workflow

1. Dataset loading
2. Data normalization
3. Train-test split
4. Logistic regression model training
5. Prediction and classification
6. Model evaluation

Workflow Diagram (Textual):

Data → Scaling → Training → Probability Estimation → Classification → Evaluation

6. Performance Analysis

Evaluation Metrics Used:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Interpretation:

- High accuracy indicates good classification

- Balanced precision and recall show robustness
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7. Hyperparameter Tuning (Logistic Regression)

Tuned Hyperparameters:

- Regularization parameter (C)
- Penalty type (L1, L2)
- Solver type

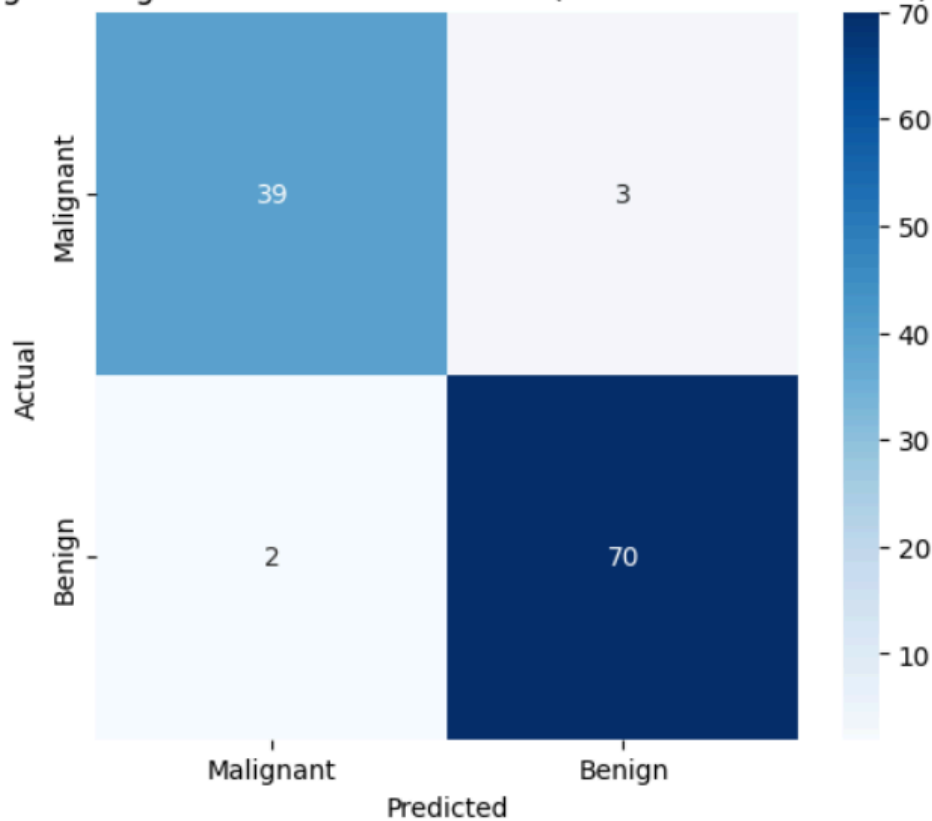
Technique Used: GridSearchCV

Impact:

- Improved classification accuracy
- Reduced overfitting

8. Output (Logistic Regression)

Logistic Regression Confusion Matrix (Breast Cancer Dataset)



Accuracy of Logistic Regression on Breast Cancer Dataset: 0.96

Conclusion

This experiment successfully demonstrated the implementation of Linear Regression and Logistic Regression on real-world datasets. Linear Regression effectively modeled continuous house price values, while Logistic Regression achieved high accuracy in binary classification. Hyperparameter tuning further enhanced model performance and generalization capability.

References

1. Scikit-learn Documentation
2. UCI Machine Learning Repository